Heating Plate Design

In semiconductor fabrication, high-temperature heating plates (up to 400°C) are critical for processes like **polyimide and polymer curing**, **photoresist hard baking**, and **thin-film annealing**.

These precision hotplates provide uniform thermal treatment to:

- Drive off solvents and cross-link chemical layers.
- Stabilize dielectrics and enhance film density before etching or deposition steps.
- Prevent substrate warping while ensuring the structural integrity of silicon wafers.

Key Specifications

A high-performance fab heating plate up to **400°C** requires rigorous semiconductor-grade specifications:

- **Thermal Uniformity:** ±1°C or tighter across the wafer to prevent thermal stress and warping.
- **Material & Purity:** High-purity anodized aluminum, silicon carbide, or nickel-plated surfaces to prevent chemical contamination and outgassing.
- **Precision Control:** Multi-zone PID controllers ensuring rapid ramp-up rates and exceptional temperature stability.
- **Wafer Handling:** Vacuum chucking or proximity-pin lifting mechanisms for flat, uniform contact without scratching.
- **Environment:** Cleanroom compatibility (ISO Class standards) with low particle generation and nitrogen purge options to prevent oxidation.

1. Design Requirements

- **It can process a 4”(~101.6mm) wafer:** I need to go with either a cylindrical design with 160mm diameter or square plate with 160mm side.
- **The plate should allow setting to a temperature range from the room temperature in the cleanroom to a maximum of 400°C:** At such high temperatures, metals begin to soften and expand. Therefore, I must be careful about these properties of the material I use.
- **The set temperature to actual temperature error should be <5°C. What’s more important in materials processing is run-to-run repeatability. The temperature from one heating run to the next one should differ by less than 0.5°C.**
    - Tool solution:
        - **Plate Metallurgy:** Constructing the plate from Aluminum Nitride (AlN) or high-grade aluminum to provide the exact ratio of thermal conductivity to thermal mass. (less hysteresis)
        - **Physical Isolation:** Building draft shields or an enclosed processing chamber into the tool frame to block ambient airflow.
    - Process solution:
        - **Standardized Thermal Baselines:** Writing the processing recipe to include a mandatory "idle soak" step (e.g., holding at setpoint for 5

minutes empty) so the tool always starts from the exact same thermodynamic state, never from a cold start.

- **The uniformity of temperature across the 4” wafer should be within +-0.5°C. That is - there should be no two points on the wafer whose temperature will vary by more than 1°C.**
- **The top surface will need to be machined flat to a tolerance of +- 10 um:** With grinding and lapping processes, this much precision can easily be obtained, but maintaining it when the material wants to thermally expand is the challenge, because if I clamp it down it will warp, especially when it softens at high temperature.

1.1 Plate Material Selection for 20-400°C Operating Range

We want a material which is:

- Highly thermally conductive to avoid hotspots and achieve temperature uniformity.
- Doesn’t warp at high temperature.
- Not too high heat capacity or else power will be wasted.
- Easy to work with.

Candidates Evaluated

- **Aluminium**: Offers great thermal conductivity (~180 W/m·K) and easy machining, but **fails at 400°C**. Standard aluminum alloys rapidly lose structural strength above 200°C and will warp and creep under mechanical stress.

- **Copper (Nickel Plated)**: Delivers exceptional thermal conductivity (~390 W/m·K) and easy machining. The nickel plating prevents oxidation at 400°C, but pure copper softens at elevated temperatures and can sag under sustained clamping stress.

- **Aluminum Bronze**: Provides good mechanical strength and oxidation resistance at 400°C, but its **lower thermal conductivity** (~50 W/m·K) makes maintaining strict temperature uniformity difficult but doable.

- **Aluminum Nitride (AlN) Ceramic**: The ultimate thermal performer with high thermal conductivity (~200 W/m·K) and zero thermal creep at 400°C. However, it is **extremely brittle** and difficult to fabricate or machine.

AI-Suggested Candidates

- **Isotropic Graphite**: Features good thermal conductivity (~100–150 W/m·K) and is **exceptionally easy to machine** into intricate heating channels. It has virtually zero thermal expansion and will not warp at 400°C (though in open air, it requires a protective coating above 400°C to prevent oxidation).

- **Molybdenum (Mo)**: **Too Expensive.** Provides solid thermal conductivity (~138 W/m·K) alongside an extremely high melting point and **outstanding creep**

resistance at 400°C. While tougher to machine than aluminum, it easily withstands heavy mechanical stress without warping.

Selected Material: In the end I chose Aluminium Bronze. However, graphite is also a viable alternative and Aluminium Nitride ceramic is ideal for a slightly higher cost.

1.2 Why a Thick 12-15mm Aluminium Bronze Plate is Needed

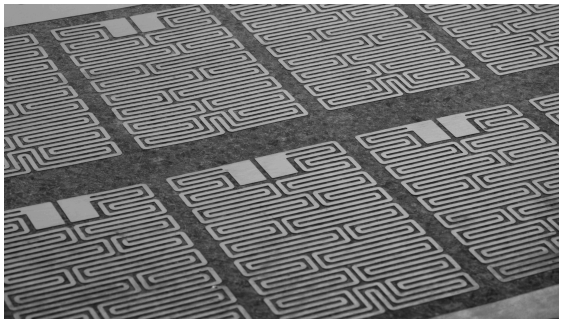
1. Lateral Heat Spreading (Erasing the Heater Pattern) Your etched stainless steel foil heater consists of high-heat metal traces separated by unheated gaps. Because Aluminum Bronze conducts heat at a moderate rate (30–70 W/mK), heat does not instantly flash sideways. As heat travels upward from a trace, it spreads outward in a cone shape. A thicker plate provides the vertical distance required for these individual "cones" of heat to overlap and blend together. If the plate is too thin, the heat reaches the top surface before blending, resulting in a thermal imprint—microscopic hot bands directly above the traces and cold bands above the gaps.

2. Thermal Mass and Load Recovery When a room-temperature 4-inch wafer is placed on a 400°C plate, it immediately pulls a massive amount of heat from the surface. A thin plate lacks thermal mass; its surface temperature will crash locally, triggering a delayed and aggressive response from the PID controller. A thick plate acts as a thermal flywheel. It holds enough stored heat energy (joules) to instantly dump heat into the wafer without significantly dropping the surface temperature, ensuring a smooth, uniform thermal ramp for the silicon.

3. Resisting Thermal Bowing (Bimetallic Effect) At 400°C, there is a temperature gradient through the plate—the bottom face touching the foil heater is hotter than the top face exposed to ambient air. Because metals expand when heated, the bottom face attempts to expand more than the top face, which creates a mechanical force that tries to bow the edges of the plate downward (doming the center upward). Increasing the thickness geometrically increases the plate's mechanical stiffness, overpowering these thermal stresses and keeping the surface perfectly flat for full wafer contact.

1.3 Heating Element

	Etched Foil Heater	**Cartridge Heater**
Thermal Uniformity	Offers superior heat distribution across the surface. Trace geometry can be tailored (e.g., adding higher wattage near edges) to compensate for perimeter heat loss, making ultra-uniform temperature profiles easier to achieve.	Acts as a concentrated line source. Requires embedding into a thick metal platen to spread heat out; hot and cold spots are common without careful simulation and placement.
Thermal Mass & Response Time	Extremely low thermal mass allows rapid ramp-up to 400°C and fast cooling when clamped to a thin plate, ideal for precise baking profiles.	Requires a massive heat spreader block, causing high thermal mass. Results in significantly longer heat-up and cool-down times.

	Etched Foil Heater	Cartridge Heater
Maximum Temperature & Materials	Standard models max out at 200°C–250°C. Reaching 400°C requires etched stainless steel foil or Inconel clamped between mica or AlN ceramic sheets.	Handles 400°C easily off-the-shelf. High-watt-density units operate at core temperatures of 700°C–800°C for high durability and reliability.
Assembly	Low-profile assembly saves space. However, fabricating custom foil and ensuring perfectly even clamping pressure across mica stacks is challenging to avoid local hot spots.	Requires deep, precision-reamed holes in a thick metal block with extremely tight tolerances to prevent overheating and failure from poor surface contact.
Image		

Conclusion

While cartridge heaters are easier to source and make Hot plates with it. But For a high-precision 4-inch wafer bake plate, **an etched stainless steel foil heater (with mica insulation) is the better choice**. Sourcing might seem difficult for an Etch Foil heater, but my father's factory can make it for us.

2. Power Calculations

(Convection is ignored. We calculate the power needed to heat the Aluminum top plate + MS pressure plate stack from 20°C to 400°C in 12 minutes, plus radiation and conduction losses at 400°C).

2.1 Mass and Thermal Capacity of the Metal Stack

1. Top Aluminum Plate (160 mm × 160 mm × 25 mm):

- Volume of Aluminum = 0.16 m × 0.16 m × 0.025 m = 0.00064 m^3
- Density of Aluminum = 2700 kg/m^3
- Mass of Aluminum Plate = 2700 kg/m^3 × 0.00064 m^3 = 1.73 kg
- Specific Heat of Aluminum (average from 20°C to 400°C) = 960 J/(kg·K)
- Heat Capacity of Aluminum Plate = 1.73 kg × 960 J/(kg·K) = 1661 J/K

2. Bottom Mild Steel (MS) Pressure Plate (160 mm × 160 mm × 10 mm):

- Volume of MS Plate = 0.16 m × 0.16 m × 0.010 m = 0.000256 m^3
- Density of Mild Steel = 7850 kg/m^3
- Mass of MS Plate = 7850 kg/m^3 × 0.000256 m^3 = 2.01 kg
- Specific Heat of Mild Steel = 510 J/(kg·K)
- Heat Capacity of MS Plate = 2.01 kg × 510 J/(kg·K) = 1025 J/K

3. Total Heat Energy Required for a 380°C Temperature Rise (from 20°C to 400°C):

- Total Energy = (1661 J/K + 1025 J/K) × 380 K
- Total Energy = 2686 J/K × 380 K = 1,020,680 Joules (approx. 1021 kJ)

2.2 Ramp-Up Power Required (12-Minute Heat-Up Time)

- Target Heat-Up Time = 12 minutes = 720 seconds
- Ramp Power = Total Energy / Time = 1,020,680 J / 720 s = **1417 W**
- *(Note: If we allow a 15-minute heat-up time, Ramp Power = 1,020,680 J / 900 s =* ***1134 W****).*

2.3 Steady-State Heat Losses at 400°C (673 K)

1. Radiation Loss from the Top Surface:

- Top Surface Area (A) = 0.16 m × 0.16 m = 0.0256 m^2
- Emissivity (e) = 0.35 (lightly oxidized ground aluminum + silicon wafer)
- Stefan-Boltzmann Constant = 5.67 × 10^{-8} $W/(m^2 \cdot K^4)$
- Hot Temp (T_hot) = 400°C = 673.15 K --> $(673.15)^4$ = 2.055 × 10^{11} K^4
- Room Temp (T_room) = 20°C = 293.15 K --> $(293.15)^4$ = 0.074 × 10^{11} K^4
- Radiation Power = 0.35 × (5.67 × 10^{-8}) × 0.0256 × (2.055 × 10^{11} - 0.074 × 10^{11})
- Radiation Power = **100.6 W**

2. Conduction Loss Down the Standoff Bolts & Bottom Insulation:

- 4 Stainless Steel M8 Standoff Pillars (k = 16 W/(m·K), total cross-section = 0.0002 m^2, length = 0.05 m):
    - Bolt Conduction Loss = (16 × 0.0002 × 380) / 0.05 = **24.3 W**
- Bottom & Side Ceramic/Calcium Silicate Insulation (k = 0.09 W/(m·K), area = 0.045 m^2, thickness = 0.025 m):
    - Insulation Conduction Loss = (0.09 × 0.045 × 380) / 0.025 = **61.6 W**

3. Total Steady-State Power to Hold 400°C:

- Steady-State Power = 100.6 W + 24.3 W + 61.6 W = **186.5 W**
- **Safety Margin at 400°C:** Once at 400°C, the heater only needs 186.5 W to maintain temperature, meaning the PID controller runs at a comfortable **12.43% duty cycle** (186.5 W / 1500 W), with 87.5% headroom.

3. Etched Foil Design

To design a 1500W etched foil heater using 0.1 mm thick SS304 foil for your 4-inch wafer hot plate, we need to balance the electrical resistance required for India's standard 230V mains with the physical constraints of a 100 mm diameter footprint.

3.1 Electrical Target (Resistance)

First, we determine the target resistance to draw 1500W at 230V AC using Ohm's Law and the Power equation:

$$P = \frac{V^2}{R} \Longrightarrow R = \frac{V^2}{P}$$

$$R = \frac{230^2}{1500} = \frac{52900}{1500} \approx 35.27\,\Omega$$

3.2 Material Constraints (Trace Dimensions)

The resistance of the foil trace is dictated by its geometry and the material's resistivity:

$$R = \rho\frac{L}{A}$$

- **Resistivity (ρ) of SS304:** At room temperature, it is approximately $7.2\times10^{-7}\,\Omega\cdot m$. *(Note: SS304 has a positive temperature coefficient, so resistance will rise slightly at your 400°C target, causing initial power to peak briefly at cold startup before stabilizing).*
- **Cross-sectional Area (A):** This is the trace width (w) multiplied by the foil thickness ($t = 0.1\,mm = 10^{-4}\,m$).
- **Trace Length (L):** The total length of the continuous etched path.

Rearranging to find the required Length-to-Width ratio:

$$35.27 = \frac{7.2\times10^{-7}\cdot L}{w\cdot10^{-4}}$$

$$\frac{L}{w} \approx 4898$$

(This means your total trace length must be roughly 4898 times its width.)

3.3 Footprint Optimization (4-inch Wafer Plate)

A standard 4-inch wafer chuck has a diameter of ~100 mm, giving a maximum usable area of about $7800\,mm^2$. Let's assume a safe active heating area of $7500\,mm^2$.

To fit the trace into this area, we arrange it in a tightly spaced serpentine pattern. The total area occupied by the trace is roughly $L \times (w + s)$, where s is the spacing (gap) between adjacent traces.

If we choose a conservative etching gap of $s = 0.5\,mm$:

$$L \times (w + 0.5) = 7500\,mm^2$$

Substitute $L = 4898 \times w$:

$$4898 \cdot w \cdot (w + 0.5) = 7500$$

$$w^2 + 0.5w = \frac{7500}{4898} \approx 1.53$$

Solving this quadratic equation yields a trace width:

$$w \approx 1.0\,mm$$

With $w = 1.0\,mm$, the required length is:

$L = 4898 \times 1.0\,mm \approx 4898\,mm$ **(or ~4.9 meters)**

3.4 Final Design Specifications

- **Material:** SS304 Foil
- **Foil Thickness:** 0.1 mm
- **Trace Width:** 1.0 mm
- **Etching Gap (Spacing):** 0.5 mm
- **Total Trace Length:** ~4898 mm (4.9 m)
- **Pattern Area:** ~7347 mm^2 (Fits well within a 100 mm diameter circle)
- **Current Draw:** ~6.52 A
- **Current Density:** $\sim 65.2\,A/mm^2$

Design Tip: Instead of a simple back-and-forth serpentine pattern, design the trace as a **dual-filar (bifilar) spiral** starting from the center out to the edge and back. This ensures more uniform heat distribution across the wafer plate and cancels out electromagnetic interference (EMI), which is critical if sensitive semiconductor test electronics will be nearby.

4. Temperature Sensor, Controller, and SSR

4.1 Temperature Sensor: PT100 (RTD) vs. Thermocouple

When targeting 400°C for processes like wafer baking, the priority is absolute accuracy and minimal drift over time.

- **Accuracy and Resolution:**

    - **PT100:** Extremely accurate. A Class A PT100 can offer an accuracy of ±0.15°C to ±0.3°C at lower ranges, and roughly ±0.9°C at 400°C. They provide a highly linear and stable output, which is critical for preventing hot/cold spots across a 4-inch wafer.
    - **Thermocouple (e.g., Type K or J):** Generally less precise. Standard Type K thermocouples have an error margin of about ±1.5°C to ±2.2°C or 0.4% (whichever is greater). They are non-linear and require cold-junction compensation, which introduces additional potential for electronic drift.
- **Operating Range & Durability:**
    - **PT100:** 400°C is pushing the upper limits for cheaper *thin-film* PT100s (which often degrade above 300°C). For sustained 400°C operation, you must specify a **wire-wound PT100**, which comfortably operates up to 600°C.
    - **Thermocouple:** Type K effortlessly handles up to 1200°C without breaking a sweat, making it more robust at extreme temperatures, though at the cost of precision.
- **Wiring:**
    - **PT100:** Uses standard copper wires. To achieve high precision, a 3-wire or 4-wire configuration is mandatory to cancel out the resistance of the lead wires.
    - **Thermocouple:** Requires specific, more expensive extension wires made of the same dissimilar metals as the sensor to avoid creating secondary parasitic thermocouples at the connection points.

Verdict for the Hot Plate: A **3-wire or 4-wire wire-wound PT100** is the superior choice. The slight increase in cost and wiring complexity is necessary to guarantee the thermal uniformity required for semiconductor processing.

4.2 Switching Element: Relay/Contactor vs. Solid State Relay (SSR)

Etched stainless steel foil heating elements have very low thermal mass. They heat up and transfer energy incredibly quickly. This characteristic completely rules out one of these switching options.

- **Switching Frequency and Control Method:**
    - **SSR (Solid State Relay):** Can switch on and off dozens of times per second. This allows a PID controller to use high-frequency PWM (Pulse Width Modulation) or time-proportioning control. As the hot plate approaches 400°C, the SSR can pulse the foil heaters with exact bursts of energy to lock the temperature in without overshooting.
    - **Relay/Contactor:** Mechanical devices with moving contacts. They can only handle slow, "bang-bang" (On/Off) control. If paired with a fast-responding foil heater, a mechanical relay will cause significant temperature ripples (hysteresis) across the plate surface, ruining the uniform bake profile.
- **Lifespan and Wear:**

    - **SSR:** Has no moving parts. Provided it is kept cool, its lifespan is virtually infinite, even when switching multiple times a second.
    - **Relay/Contactor:** Physical contacts arc and degrade every time they open or close. If a PID controller attempts to pulse a mechanical contactor to hold 400°C, the contactor will destroy itself mechanically or weld shut in a very short time.
- **Thermal Management:**
    - **SSR:** The primary downside is that SSRs generate heat (roughly 1 to 1.5 watts of heat per amp switched). For a high-wattage wafer hot plate, the SSR *must* be mounted to an appropriately sized aluminum heatsink with thermal compound, and potentially actively cooled with a small fan.
    - **Relay/Contactor:** Generates negligible heat at the switching site and requires no heatsinking.

Verdict for the Hot Plate: An **SSR paired with a high-quality PID controller** is absolutely mandatory. A mechanical relay cannot switch fast enough to maintain a flat, uniform temperature profile on low-mass etched foil heaters, and would quickly fail from the rapid cycling. Ensure you use a Zero-Crossing SSR to minimize electrical noise (EMI) during switching.



We will be using the Multipsan UTC-413 temperature controller. It cost less than Rs1000 and it includes temperature sensor.

5. Bill of Materials (BOM)

All parts below can be sourced in Mumbai (Lohar Chawl, Grant Road, Metal Market Kalbadevi) or online via Robu.in / IndiaMART, keeping the total cost well within target.

#	Component	Specification & Purpose	Qty	Est. Cost (INR)

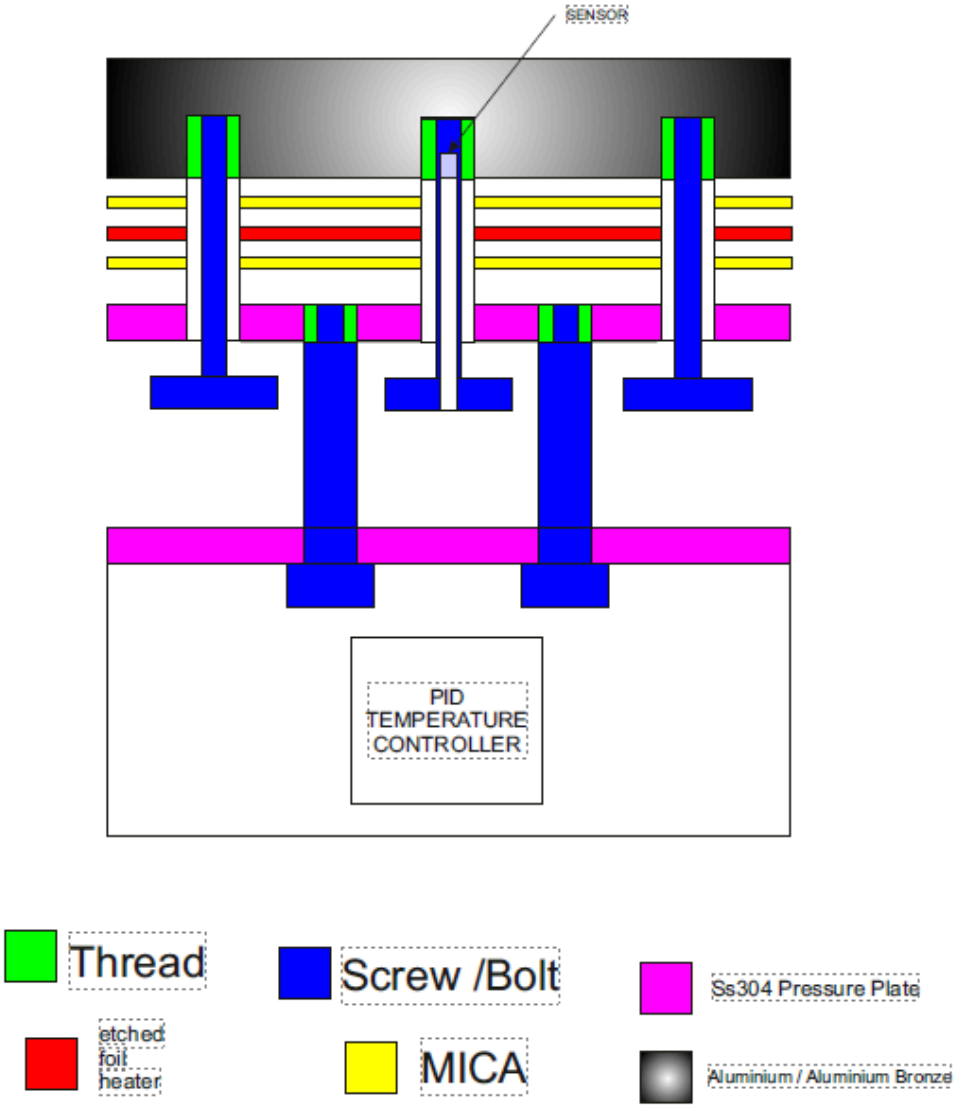
1	Top Heating Plate Stock	Aluminum Bronze (C61400 or C95400 alloy) Square Block, 165 × 165 × 12–15 mm (~2.5–3.1 kg)	1	2,400
2	MS Pressure Plate & Base Plate	Mild Steel (EN8 / IS 2062) 165 × 165 × 12 mm + 165 × 165 × 6 mm lower plate	1 Set	650
3	Stress-Relief & Surface Grinding	425°C furnace anneal + precision wet surface grinding to ±10 µm	1 Job	3,000
4	Custom Etched SS304 Foil Heater	Custom Etched SS304 Foil Heater (1500W, 0.1 mm thickness, 1.0 mm trace width, 0.5 mm etching gap)	1	4,500
5	High-Temp Fasteners & Springs	High-temperature threading bolts/screws, Belleville disc springs	1 Lot	1,150
6	Primary Temperature Sensor	Wire-wound PT100 RTD (3-wire or 4-wire, ceramic core, high-temp rated), Ø4 mm × 50 mm SS316 probe	1	1,800
7	PID Temperature Controller	Multispan UTC-413 PID Temperature Controller, 0.1°C display resolution, SSR drive output	1	2,800

8	Solid State Relay (SSR) + Heatsink	Zero-Crossing Solid State Relay (SSR) + Aluminum Heatsink for the SSR	1	950
9	Thermal Paste	Boron Nitride (BN) High-Temp Thermal Paste (rated to 800°C, electrically insulating)	1 Tube	1,400
10	Insulation	High-Temperature Mica Sheet	1 Sheet	950
11	Over-Temperature Safety Cutout	500°C Mechanical Capillary Thermostat (wired in series with AC Live for fail-safe cutoff)	1	650
12	Control Box & Electricals	Powder-coated sheet metal enclosure, 10A DP MCB, ceramic terminal block, fiberglass wire, 16A power cord	1 Lot	2,400
TOTAL				**INR 22,550**

6. Assembly Instructions

1. **Platen Preparation:** CNC machine the aluminum bronze top plate to specification, ensuring the top surface is ground flat to the $+- 10\mu m$ tolerance. Drill the necessary channels for the center-mounted sensor and mounting threads.
2. **Heater Stack Assembly:** Invert the bronze plate. Lay down the first high-temperature mica sheet, followed by the etched SS304 foil heater, and cover it with the second mica sheet to completely isolate the electrical element.

3. **Clamping:** Position the SS304 pressure plate over the lower mica sheet. Fasten the assembly using bolts into the threaded bronze plate. Torque the bolts in a star pattern to ensure perfectly even clamping pressure across the entire mica stack, which is critical to prevent the foil from developing local hot spots.
4. **Sensor Integration:** Insert the wire-wound PT100 sensor into the central well of the bronze plate, ensuring direct physical contact with the thermal mass.
5. **Electronics Wiring:** Mount the SSR to the aluminum heatsink to manage the 1 to 1.5 watts of heat generated per amp switched. Wire the PT100 sensor to the Multispan UTC-413 PID controller, and connect the PID output to the SSR to control the 230V mains supply to the foil heater.



(exploded top-down view of the assembly)